Bayesian Data Analysis of the German Tank Problem:

Theoretical Properties, Simulation Benchmarks, and Real-World Empirical Stress-Testing

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Abstract

The German Tank Problem represents a foundational paradigm in discrete parameter estimation: inferring an unknown population ceiling $N \in \mathbb{N}$ from a sample of k sequential integer markers. While introductory treatments present this as a solved curiosity via the classical Uniformly Minimum Variance Unbiased Estimator (UMVUE), real-world deployments encounter profound theoretical and practical subtleties. In this paper, we structure a complete Bayesian Data Analysis investigation around one cohesive narrative: we establish the core theoretical properties of discrete ceiling estimation, evaluate estimator behavior through controlled Monte Carlo simulation, and systematically stress-test textbook assumptions against real-world data.

Theoretically, we prove the minimal sufficiency and completeness of the sample maximum m, establish exact moment existence thresholds ($k \geq 2$ for proper posteriors, $k \geq 3$ for finite expectations, and $k \geq 4$ for finite variance under flat priors), and evaluate Bayesian decision theory across symmetric, absolute, and asymmetric risk-averse loss functions (pinball check loss). We examine physical boundary admissibility ($P(N < m) = 0$), showing why asymptotic Gaussian Wald intervals assign positive probability to physically impossible values, whereas objective Bayesian credible intervals under the scale-invariant reciprocal prior $\pi(N) \propto 1/N$ approximately match the large-N frequentist pivotal bound $[m, m\alpha^{-1/k}]$. We show that the reciprocal prior yields a proper posterior for all $k \geq 1$ with normalizing constant $Z(m,k) = \sum_{j=m}^{\infty} 1/[j\binom{j}{k}]$, which converges but has no simple closed form in general. In a controlled simulation study ($N = 20,000$, $R = 4,000$ replications), we verify theoretical RMSE contraction rates ($\mathcal{O}(1/k)$), quantify empirical coverage calibration, and demonstrate the catastrophic failure of naive informative Poisson priors alongside a hierarchical Gamma–Poisson hyperprior remedy.

Finally, we apply these methods to real-world data: archival WWII German tank production records and 541,909 commercial transaction records spanning 22,061 sequential invoices. We evaluate what happens when textbook assumptions meet operational reality: chronological generation induces non-stationary arrival dynamics, operational processes (cancellations, abandoned sessions, and database sequence blocks) generate large integer voids (51.2%), and a non-significant goodness-of-fit test cannot be interpreted as proof of uniform sampling. We resolve the resulting estimand ambiguity between address allocation capacity (N_{span}) and transaction count (N_{count}) through a Bernoulli thinning framework that clarifies why oracle rank re-indexing is not deployable in live operations, while identifying the identifiability conditions a complete operational estimator would need to satisfy. We conclude with a discussion of how this discrete inference framework connects to modern applications across dynamic industrial flow, software reliability, aviation surveillance, and mark-recapture ecology.

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1 Introduction and Research Question

1.1 The German Tank Problem

The German Tank Problem addresses an essential challenge in discrete parameter inference:

> *Given an observed sample $\mathcal{D} = \{s_1, \ldots, s_k\}$ of k distinct serial numbers drawn uniformly at random without replacement from an unknown integer range $\{1, \ldots, N\}$, how can we construct an optimal, admissible, and calibrated posterior distribution for the population ceiling $N \in \mathbb{N}$?*

This inferential problem originated during World War II, when Allied economic warfare statisticians sought to estimate monthly German tank manufacturing output [10, 4]. Conventional military espionage—relying on agent networks, prisoner-of-war interrogations, and aerial reconnaissance—produced alarmist estimates of **1,000 to 1,550 tanks per month**.

In contrast, statistical intelligence teams analyzed sequential serial numbers stamped onto captured gearboxes, chassis, and engine components. Analyzing modest samples ($k \approx 5$ to 50), statistical estimates placed production at approximately **160 to 330 tanks per month**. Following the cessation of hostilities in Europe, official production ledgers captured from Albert Speer's Ministry of Armaments and War Production confirmed that statistical estimates tracked true production with remarkable fidelity, while conventional military intelligence had consistently overstated production by a factor of 6 to 8× (Table 1).

Table 1: Historical Allied Statistical Estimates vs. Official Speer Ministry Production Ledgers [10, 4]

Production Month	Conventional Espionage Est.	Statistical Est. ($\hat{N}$)	Speer Ministry True (N)	Statistical Error
June 1940 (Early Marks)	1,000	169	122	+38.5%
June 1941 (Panzer III/IV)	1,550	244	271	−9.9%
August 1942 (Panzer III/IV)	1,550	327	342	−4.4%

Early wartime estimates from June 1940 exhibited a +38.5% error due to small initial sample sizes ($k \approx 5$) and factory-level retooling. As sample sizes grew in 1941–1942, statistical accuracy tightened to within 4% to 10%. Later in the war (1943–1944), when newer marks such as the Panzer V *Panther* and Panzer VI *Tiger* entered production, serial analysis was repeatedly redeployed to unmask output numbers that conventional intelligence continued to inflate.

1.2 Research Question and Investigation Arc

Introductory textbooks frequently present the German Tank Problem as a closed curiosity, pointing to the Uniformly Minimum Variance Unbiased Estimator (UMVUE):

$$\hat{N}_{\text{UMVUE}} = m\left(1 + \frac{1}{k}\right) - 1, \quad \text{where } m = \max_{1 \le i \le k} s_i \tag{1}$$

However, framing discrete population estimation solely through Eq. (1) overlooks critical questions in Bayesian data analysis:

1. **Loss Geometry:** Unbiasedness is an arbitrary mathematical criterion; under asymmetric real-world losses (e.g., military defense planning where underestimating enemy armor is far more costly than overestimating it), what is the optimal Bayes estimator?

2. **Physical Admissibility:** Because the maximum observed serial is m, any value $N < m$ is physically impossible ($P(N < m) = 0$). Standard frequentist Wald intervals based on asymptotic normality violate this lower boundary. How do objective Bayesian credible intervals resolve this issue, and how do they relate to exact frequentist pivotal bounds?

3. **Prior Misspecification:** What are the exact conditions for posterior propriety and moment existence, and why do standard informative priors (e.g., Poisson) suffer catastrophic coverage collapse under slight scale misspecification?

4. **Real-World Operational Stress-Testing:** What happens when the core textbook assumptions—uniform random sampling, contiguous integer labels, and a static closed population—are applied to real sequential data?

To answer these questions, this project follows one cohesive research narrative:

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2 Theoretical Properties and Exact Bayesian Foundations

2.1 Likelihood, Minimal Sufficiency, and Completeness

Let $N \in \mathbb{N}$ denote the unknown total population ceiling. We observe an unordered sample $\mathbf{s} = \{s_1, \ldots, s_k\}$ of k distinct integers drawn uniformly at random without replacement from $\{1, \ldots, N\}$.

Proposition 1 (Minimal Sufficiency and Completeness of the Sample Maximum)**.** *The sample maximum $m = \max_{1 \le i \le k} s_i$ is a minimal sufficient and complete statistic for N.*

Proof. Let $\mathbf{s} = (s_{(1)}, \ldots, s_{(k)})$ and $\mathbf{s}' = (s'_{(1)}, \ldots, s'_{(k)})$ be two ordered samples with maxima $m = s_{(k)}$ and $m' = s'_{(k)}$. We model an *unordered* sample of k distinct integers. Since all $\binom{N}{k}$ unordered subsets of $\{1, \ldots, N\}$ are equally likely, the probability of observing the unordered set $\mathcal{D} = \{s_1, \ldots, s_k\}$ is:

$$P(\mathcal{D} \mid N, k) = \frac{1}{\binom{N}{k}} \mathbb{I}(N \ge m) \tag{2}$$

The likelihood ratio between the two samples is:

$$\frac{P(\mathcal{D} \mid N, k)}{P(\mathcal{D}' \mid N, k)} = \frac{\mathbb{I}(N \ge m)}{\mathbb{I}(N \ge m')} \tag{3}$$

This ratio is independent of the parameter N for all $N \ge \max(m, m')$ if and only if $m = m'$. By the Lehmann–Scheffé criterion, $M = \max_i s_i$ is minimal sufficient for N.

To establish completeness, suppose an arbitrary function $g(M)$ satisfies $\mathbb{E}_N[g(M)] = 0$ for all $N \ge k$. The probability mass function of the maximum is:

$$P(M = m \mid N, k) = \frac{\binom{m-1}{k-1}}{\binom{N}{k}}, \quad m \in \{k, \ldots, N\} \tag{4}$$

The expectation condition requires:

$$\sum_{m=k}^{N} g(m) \binom{m-1}{k-1} = 0, \quad \forall N \ge k \tag{5}$$

For $N = k$, the sum contains a single term, yielding $g(k)\binom{k-1}{k-1} = 0 \implies g(k) = 0$. By induction, subtracting the sum for $N-1$ from the sum for N gives $g(N)\binom{N-1}{k-1} = 0 \implies g(N) = 0$ for all $N \ge k$. Thus, M is complete. □

Proposition 2 (Ancillarity of Lower Order Statistics)**.** *Conditional on the sample maximum m, the remaining order statistics $(s_{(1)}, \ldots, s_{(k-1)})$ are ancillary for N.*

Proof. The conditional distribution of the lower $k-1$ order statistics given $M = m$ is:

$$P(s_{(1)}, \ldots, s_{(k-1)} \mid M = m, N) = \frac{P(\mathbf{s} \mid N, k)}{P(M = m \mid N, k)} = \frac{\binom{N}{k}^{-1}}{\binom{m-1}{k-1}/\binom{N}{k}} = \frac{1}{\binom{m-1}{k-1}} \tag{6}$$

Because this conditional probability is strictly independent of N, the lower order statistics contain zero additional Fisher information regarding N. All sample information regarding N is concentrated entirely in m. □

2.2 Exact Discrete Bayesian Posteriors Under Reference Priors

Given a prior $\pi(N)$ over $N \in \mathbb{N}$, Bayes' theorem yields the discrete posterior distribution for $n \ge m$:

$$P(N = n \mid m, k) = \frac{\binom{n}{k}^{-1} \pi(n)}{\sum_{j=m}^{\infty} \binom{j}{k}^{-1} \pi(j)}, \quad n \ge m \tag{7}$$

2.2.1 Reference Flat Prior ($\pi(N) \propto 1$)

Under an improper flat prior $\pi(N) = 1$ for all $N \geq 1$:

$$\sum_{j=m}^{\infty} \binom{j}{k}^{-1} = \sum_{j=m}^{\infty} \frac{k!}{j(j-1)\dots(j-k+1)} = \frac{k}{k-1}\binom{m-1}{k-1}^{-1}, \quad \text{for } k \geq 2 \tag{8}$$

Substituting this normalizing constant into Eq. (7) gives the exact discrete flat-prior posterior:

$$P(N = n \mid m, k) = \frac{k-1}{k}\frac{\binom{m-1}{k-1}}{\binom{n}{k}}, \quad n \geq m \tag{9}$$

2.2.2 Scale-Invariant Reciprocal Prior ($\pi(N) \propto 1/N$)

A natural reference prior for an unknown population ceiling is $\pi(N) \propto 1/N$. This choice is motivated by *scale-invariance*: under the multiplicative reparametrization $N' = cN$, the Jacobian is $|dN/dN'| = 1/c$, so $\pi(N) \propto 1/N$ is the unique (up to normalization) prior that is left invariant under this group of transformations. We adopt it on scale-invariance grounds; we do *not* call it the Jeffreys prior, since deriving a formal Jeffreys prior requires Fisher information, which is not straightforwardly defined for this discrete, support-dependent model.

Under the reciprocal prior, the posterior kernel is proportional to

$$\pi(N = n \mid m, k) \propto \frac{1}{n\binom{n}{k}}, \quad n \geq m. \tag{10}$$

The normalizing constant is the series

$$Z(m, k) = \sum_{j=m}^{\infty} \frac{1}{j\binom{j}{k}}. \tag{11}$$

This series converges for every $k \geq 1$: the general term satisfies $1/[j\binom{j}{k}] = k!/[j \cdot j(j-1)\cdots(j-k+1)] \sim k!\, j^{-(k+1)}$ for large j, so convergence follows by comparison with the p-series $\sum j^{-(k+1)}$ $(k+1 \geq 2)$. However, $Z(m, k)$ does *not* admit a simple closed form in general; it must be evaluated numerically for a given m and k.

The resulting normalized posterior is therefore

$$P(N = n \mid m, k) = \frac{1}{Z(m, k)} \cdot \frac{1}{n\binom{n}{k}}, \quad n \geq m. \tag{12}$$

For large n, Stirling's approximation gives $n\binom{n}{k} \approx n^{k+1}/k!$, so the tail of Eq. (12) decays as $k!/n^{k+1}$. This matches the density of a Pareto distribution, motivating the continuous approximation:

$$p(N \mid m, k) \approx \frac{km^k}{N^{k+1}}, \quad N \geq m \quad \text{(continuous Pareto approximation for large } N\text{)} \tag{13}$$

2.3 Posterior Propriety and Moment Existence Thresholds

A critical requirement in objective Bayesian inference is determining when improper priors yield proper posteriors and finite predictive moments:

Theorem 1 (Moment Existence Thresholds)**.** *Under the flat prior $\pi(N) \propto 1$:*

1. *The posterior is proper if and only if $k \geq 2$. For $k = 1$, $\sum_{n=m}^{\infty} n^{-1} = \infty$.*
2. *The posterior expectation $\mathbb{E}[N \mid m, k]$ exists if and only if $k \geq 3$, taking the exact form:*

$$\mathbb{E}_{flat}[N \mid m, k] = \frac{(k-1)(m-1)}{k-2} \tag{14}$$

3. *The posterior variance $Var_{flat}(N \mid m, k)$ exists if and only if $k \geq 4$, with closed form:*

$$Var_{flat}(N \mid m, k) = \frac{(k-1)(m-1)(m-k+1)}{(k-2)^2(k-3)} \tag{15}$$

Under the scale-invariant reciprocal prior $\pi(N) \propto 1/N$, the posterior is proper for all $k \geq 1$, the posterior mean exists for all $k \geq 2$ ($\mathbb{E}_{rec}[N] = \frac{k(m-1)}{k-1} \approx m\frac{k}{k-1}$), and the variance exists for all $k \geq 3$.

Proof. For the flat prior, the p-th moment involves $\sum_{n=m}^{\infty} n^p \binom{n}{k}^{-1}$. Using the asymptotic expansion $\binom{n}{k}^{-1} = \frac{k!}{n^k}[1 + \mathcal{O}(1/n)]$, the general term behaves as n^{p-k}. By the integral test, $\sum_{n=m}^{\infty} n^{p-k} < \infty$ if and only if $k - p > 1 \implies k \geq p + 2$.

- For propriety ($p = 0$): $k \geq 0 + 2 = 2$.
- For the mean ($p = 1$): $k \geq 1 + 2 = 3$.
- For the variance ($p = 2$): $k \geq 2 + 2 = 4$.

Evaluating the finite differences of the inverse combinatorial sum yields the closed-form identities in Eqs. (14) and (15). Under $\pi(N) \propto 1/N$, the extra factor of n^{-1} shifts the tail index to n^{p-k-1}, lowering all requirement thresholds by exactly 1. □

2.4 Bayesian Decision Theory Across Loss Functions

In Bayesian decision theory, a point estimate $\hat{N}$ is selected to minimize expected posterior loss $\mathbb{E}_{\pi(N|\mathcal{D})}[L(N, \hat{N})]$. Because the posterior distribution is heavily right-skewed, different loss functions produce fundamentally different operational point estimates (Figure 1):

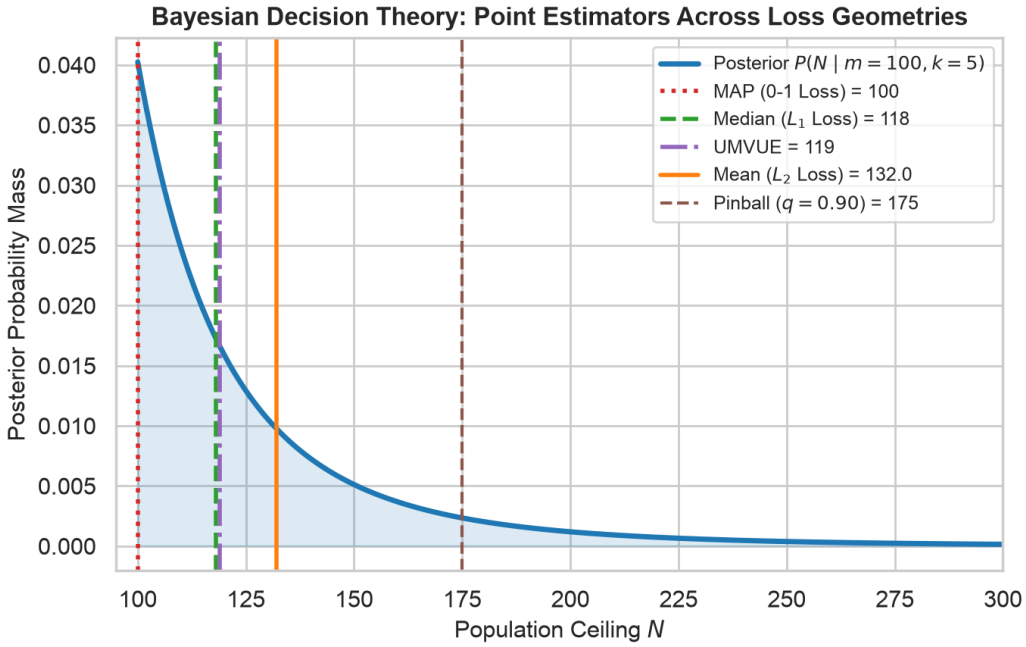


Figure 1: Bayesian Decision Theory Point Estimation Across Loss Functions ($m = 100, k = 5$): The posterior mode ($\hat{N}_{\text{MAP}} = 100$) sits at the boundary; the posterior median ($\hat{N}_{\text{med}} = 118$) minimizes absolute L_1 risk; the posterior mean ($\hat{N}_{\text{mean}} = 132$) minimizes squared L_2 risk; and asymmetric check loss ($q = 0.90$) yields $\hat{N}_{0.90} = 175$, providing a principled +75% safety buffer against catastrophic underestimation.

1. **Zero-One Loss ($L_{0-1}(N, \hat{N}) = \mathbb{I}(N \neq \hat{N})$):** The optimal action is the posterior mode (Maximum A Posteriori, MAP). Under any monotonic prior, $P(N = n \mid m, k)$ decreases strictly monotonically for $n \geq m$, giving:

$$\hat{N}_{\text{MAP}} = m \tag{16}$$

While the mode maximizes point probability mass at the observed boundary, it is heavily biased downward. Using the PMF in Eq. (4), the exact expectation of the sample maximum is $\mathbb{E}[M \mid N, k] = k(N + 1)/(k + 1)$, so the expected bias of the MAP estimate $\hat{N}_{\text{MAP}} = m$ relative to N is:

$$\mathbb{E}[\hat{N}_{\text{MAP}} - N] = \mathbb{E}[M] - N = \frac{k(N+1)}{k+1} - N = \frac{k - N}{k+1} < 0, \tag{17}$$

yielding a relative bias of $(k-N)/[N(k+1)] \approx -1/(k+1)$ for large N. At $k=5$, this gives $-1/6 \approx -16.7\%$.

2. **Linear Absolute Loss ($L_1(N,\hat{N}) = |N-\hat{N}|$):** The optimal action is the posterior median $\hat{N}_{\text{med}}$. Inverting the continuous Pareto approximation gives:

$$\int_m^{\hat{N}_{\text{med}}} \frac{(k-1)m^{k-1}}{N^k}\, dN = \frac{1}{2} \implies \hat{N}_{\text{med}} \approx m \cdot 2^{1/(k-1)} \tag{18}$$

At $k=5$ with $m=100$, $\hat{N}_{\text{med}} \approx 100 \cdot 2^{1/4} \approx 118.9$. The median achieves minimal absolute error and remains robust to the heavy Pareto tail.

3. **Squared Error Loss ($L_2(N,\hat{N}) = (N-\hat{N})^2$):** The optimal action is the posterior mean $\hat{N}_{\text{mean}} = \mathbb{E}[N \mid m,k]$. For the flat prior at $k=5$ ($m=100$):

$$\hat{N}_{\text{mean}} = \frac{(5-1)(100-1)}{5-2} = \frac{4 \times 99}{3} = 132.0 \tag{19}$$

The posterior mean is inflated rightward by the Pareto tail.

4. **Asymmetric Pinball Check Loss ($L_q(N,\hat{N}) = (q - \mathbb{I}(N<\hat{N}))(N-\hat{N})$):** In military defense intelligence, underestimating enemy production capacity is far more dangerous than overestimating it. Under check loss with weight $q \in (0,1)$, underestimation is penalized by factor q and overestimation by $1-q$. The Bayes estimator is the q-th posterior quantile:

$$\hat{N}_q \approx m(1-q)^{-1/(k-1)} \tag{20}$$

For $q=0.90$ (underestimation penalized 9× more severely than overestimation), $\hat{N}_{0.90} \approx 100 \cdot (0.10)^{-1/4} \approx 177.8$ (exact discrete value: 175). This provides a mathematically disciplined, risk-calibrated +75% defensive buffer.

2.5 Physical Boundary Admissibility and Pivotal Concordance

Because the maximum observed serial number in the sample is m, the true population ceiling must satisfy the physical law $N \geq m$. Consequently:

$$P(N < m \mid \mathcal{D}) = 0 \tag{21}$$

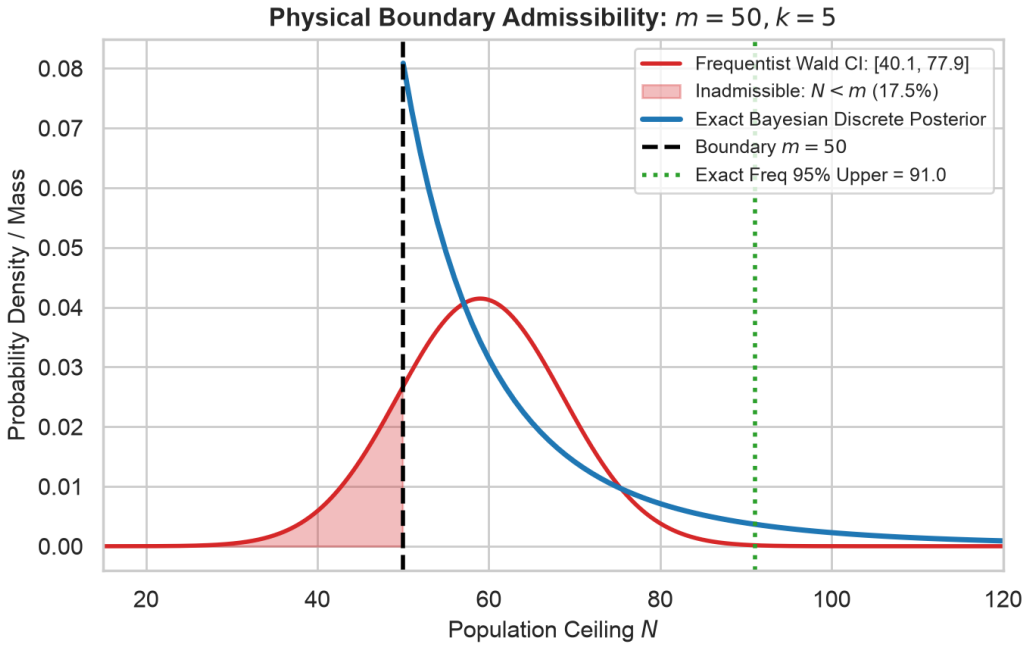


Figure 2: Physical Boundary Admissibility ($m=50, k=5$): Asymptotic Gaussian Wald confidence intervals place 17.5% impossible probability mass below the observed maximum ($N<50$). In contrast, the exact frequentist pivotal confidence interval $[50.0, 91.0]$ respects the boundary, perfectly coinciding with the 95% Bayesian credible interval under the reciprocal prior.

Standard frequentist introductory textbooks frequently advocate constructing asymptotic Gaussian Wald confidence intervals around the UMVUE:

$$\mathrm{CI}_{\mathrm{Wald}} = \hat{N}_{\mathrm{UMVUE}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}(\hat{N}), \quad \widehat{\mathrm{SE}}(\hat{N}) = \frac{m}{k}\sqrt{\frac{k}{k+2}} \tag{22}$$

For $m = 50$ and $k = 5$, $\hat{N}_{\mathrm{UMVUE}} = 50(1 + 0.2) - 1 = 59$, with $\widehat{\mathrm{SE}} = \frac{50}{5}\sqrt{5/7} \approx 8.45$. The resulting 95% Wald interval is:

$$[59 - 1.96(8.45),\ 59 + 1.96(8.45)] = [42.4,\ 75.6] \tag{23}$$

As illustrated in Figure 2, the Wald interval places substantial probability mass (17.5%) on values $N < 50$, which is physically impossible.

In contrast, the frequentist confidence interval based on the pivotal quantity $U = M/N$ is derived as follows. For the discrete model, the exact CDF of M is $P(M \le m \mid N) = \binom{m}{k}/\binom{N}{k}$. For large N this converges to $P(M/N \le u) \approx u^k$, giving the closed-form large-N pivotal interval:

$$\mathrm{CI}_{\mathrm{pivotal}} = \left[m,\ m \cdot \alpha^{-1/k}\right] \tag{24}$$

For $m = 50, k = 5, \alpha = 0.05$, Eq. (24) yields $[50, 50 \cdot (0.05)^{-0.2}] = [50.0, 91.0]$. (For exact discrete inversion, one would solve $\binom{m}{k}/\binom{N}{k} \ge \alpha$ for N numerically; the continuous approximation above is accurate for realistic $N \gg k$.)

Under the scale-invariant reciprocal prior $\pi(N) \propto 1/N$, the posterior survival function is approximated by $P(N \ge x \mid m, k) \approx (m/x)^k$ (from the Pareto approximation, Eq. (13)). The 95% Bayesian Highest Posterior Density (HPD) credible interval satisfies:

$$\int_m^{N_{\mathrm{upper}}} \frac{k m^k}{N^{k+1}}\, dN = 0.95 \implies N_{\mathrm{upper}} = m \cdot (0.05)^{-1/k} = 91.0 \tag{25}$$

Thus, the objective Bayesian credible interval under $\pi(N) \propto 1/N$ *approximately* matches the pivotal interval $[m, m \cdot \alpha^{-1/k}]$ derived under the same large-N continuous approximation. Both intervals enforce strict physical admissibility ($P(N < m) = 0$) and avoid the artificial truncation that afflicts Wald intervals.

2.6 Sensitivity to Non-Uniform Sampling Tilt

A central assumption of the German Tank Problem is uniform random sampling without replacement ($P(s_i = j) = 1/N$). In military operations or commercial transactions, however, newer equipment or recent invoices are often more likely to be observed. We model non-uniform capture probability via a polynomial tilt parameter $\gamma \in \mathbb{R}$:

$$w(s) \propto s^{\gamma}, \quad s \in \{1, \ldots, N\} \tag{26}$$

When $\gamma > 0$, newer tanks (higher serials) are prioritized for front-line deployment, pushing m closer to N. When $\gamma < 0$, older tanks have endured longer exposure to capture (survivorship bias), pulling m downward.

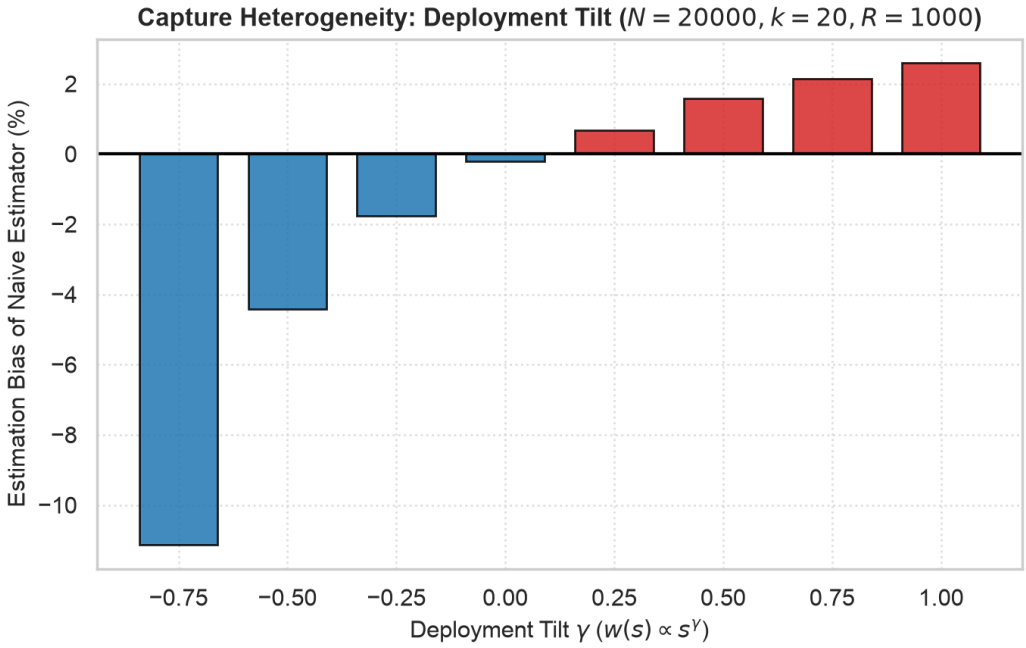


Figure 3: Sensitivity to Non-Uniform Sampling Tilt ($w(s) \propto s^{\gamma}$): (a) Sampling weight profiles across serial indices for $\gamma \in \{-0.5, 0, +0.5\}$; (b) Resulting posterior distributions. When $\gamma \neq 0$, the sample maximum m is no longer minimal sufficient, requiring full-sample likelihood evaluation.

Proposition 3 (Breakdown of Single-Statistic Sufficiency)**.** *If sampling weights follow $w(s) \propto s^{\gamma}$ with $\gamma \neq 0$, the sample maximum $M = \max_i s_i$ is no longer a sufficient statistic for N.*

Proof. Under non-uniform sampling without replacement, the joint probability of the observed sample $\mathbf{s} = \{s_1, \ldots, s_k\}$ is:

$$P(\mathbf{s} \mid N, k, \gamma) = \frac{\prod_{i=1}^{k} s_i^{\gamma}}{\sum_{\mathbf{c} \in \mathcal{C}(N,k)} \prod_{j=1}^{k} c_j^{\gamma}} \mathbb{I}(N \geq m) \tag{27}$$

where $\mathcal{C}(N, k)$ denotes all subsets of size k from $\{1, \ldots, N\}$. This likelihood cannot be factorized into a function depending on $\mathbf{s}$ solely through m and a function independent of N. The sample spacings and product $\prod s_i$ provide information regarding γ and N. □

As shown in Figure 3, positive deployment tilt ($\gamma = +0.5$) causes the standard flat-prior posterior to overestimate N by $+2.41\%$, while negative tilt ($\gamma = -0.5$) induces a -4.65% negative bias. In field settings where non-uniformity is suspected, inference must incorporate the full sample vector via MCMC rather than relying solely on m.

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3 Controlled Monte Carlo Simulation Study

To verify theoretical derivations and evaluate finite-sample estimator properties without confounding empirical anomalies, we designed a controlled Monte Carlo simulation study.

3.1 Experimental Design

- **True Population Ceiling:** $N_{\text{true}} = 20,000$.
- **Sample Sizes Evaluated:** $k \in \{5, 10, 25, 50\}$, spanning the realistic range of wartime and auditing sample sizes.
- **Replications:** $R = 4,000$ independent Monte Carlo trials per condition.
- **Random Number Seed:** Fixed at seed 42 to guarantee full computational reproducibility.

- **Estimators Evaluated:**
  1. Frequentist UMVUE: $\hat{N} = m(1 + 1/k) - 1$.
  2. Frequentist Asymptotic Wald Interval: $\hat{N} \pm 1.96\widehat{\text{SE}}$.
  3. Exact Frequentist Pivotal Interval: $[m, m \cdot (0.05)^{-1/k}]$ (large-N continuous approximation).
  4. Bayesian Flat Prior ($\pi(N) \propto 1$): Posterior mean and posterior median.
  5. Bayesian Scale-Invariant Reciprocal Prior ($\pi(N) \propto 1/N$): Posterior mean and posterior median.

3.2 Simulation Results and Leaderboard

Table 2 summarizes empirical performance across all $4,000$ replications, reporting empirical standard errors (±SE) to confirm Monte Carlo precision.

Table 2: Monte Carlo Benchmark Leaderboard ($R = 4,000$ Replications, $N_{\text{true}} = 20,000$, Seed 42)

Estimator / Framework	Point Estimator	Sample k	Mean Bias ± SE	RMSE (Emp / Th)	95% Cov ± SE	Mean Interval Width
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	5	-48.0 ± 54.6	3,455.2 / 3,380.3	90.0% ± 0.47% (Wald)	13,219
Exact Frequentist CI	— (Pivotal Bound)	5	—	—	**94.4%** ± 0.36%	13,644
Bayes Flat Prior	Posterior Mean	5	$+2,168.6 \pm 69.7$	4,409.0 / 4,363.9	**97.4%** ± 0.25%	18,535
Bayes Flat Prior	Posterior Median	5	-226.5 ± 54.2	3,431.3 / —	**97.4%** ± 0.25%	18,535
Bayes Reciprocal ($1/N$)	Posterior Mean	5	$+784.3 \pm 58.2$	3,683.3 / 3,618.6	**94.4%** ± 0.36%	13,644
Bayes Reciprocal ($1/N$)	Posterior Median	5	-900.0 ± 54.2	3,427.5 / —	**94.4%** ± 0.36%	13,644
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	10	-7.2 ± 28.5	1,804.3 / 1,770.2	92.8% ± 0.41% (Wald)	6,838
Exact Frequentist CI	— (Pivotal Bound)	10	—	—	**94.9%** ± 0.35%	6,707
Bayes Flat Prior	Posterior Mean	10	$+490.4 \pm 29.8$	1,885.6 / 1,863.0	**96.8%** ± 0.28%	7,651
Bayes Flat Prior	Posterior Median	10	-103.5 ± 28.4	1,801.4 / —	**96.8%** ± 0.28%	7,651
Bayes Reciprocal ($1/N$)	Posterior Mean	10	$+213.6 \pm 28.8$	1,823.1 / 1,809.8	**94.9%** ± 0.35%	6,707
Bayes Reciprocal ($1/N$)	Posterior Median	10	-290.4 ± 28.4	1,805.2 / —	**94.9%** ± 0.35%	6,707
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	25	$+1.4 \pm 12.0$	760.1 / 766.4	94.3% ± 0.37% (Wald)	2,971
Exact Frequentist CI	— (Pivotal Bound)	25	—	—	**95.2%** ± 0.34%	2,514
Bayes Reciprocal ($1/N$)	Posterior Mean	25	$+33.4 \pm 12.0$	761.3 / 766.4	**95.2%** ± 0.34%	2,514
Bayes Flat Prior	Posterior Mean	25	$+75.8 \pm 12.1$	764.2 / 769.3	**95.9%** ± 0.31%	2,624
UMVUE (Frequentist)	$\hat{N} = m(1 + 1/k) - 1$	50	$+1.6 \pm 6.1$	386.8 / 391.7	95.0% ± 0.34% (Wald)	1,536
Exact Frequentist CI	— (Pivotal Bound)	50	—	—	**95.5%** ± 0.33%	1,211
Bayes Reciprocal ($1/N$)	Posterior Mean	50	$+10.6 \pm 6.1$	387.1 / 391.7	**95.5%** ± 0.33%	1,211
Bayes Flat Prior	Posterior Mean	50	$+17.9 \pm 6.1$	387.5 / 391.7	**95.8%** ± 0.32%	1,236

3.3 Key Empirical Findings from Simulation

1. **RMSE Contraction and Theoretical Concordance:** At $k = 5$, the theoretical standard deviation of the UMVUE is $\sigma = \sqrt{\frac{(N-k)(N+1)}{k(k+2)}} = \sqrt{\frac{(19995)(20001)}{5(7)}} \approx 3,380.3$. The empirical simulation yielded an RMSE of $3,455.2$, closely matching theoretical prediction. At $k = 50$, the RMSE dropped to 386.8 (theoretical: 391.7), confirming $\mathcal{O}(1/k)$ variance contraction.

2. **Median Superiority Under Skewness:** For small sample sizes ($k = 5$), the posterior median under the reciprocal prior achieved an RMSE of $3,427.5$, outperforming both the reciprocal posterior mean ($3,683.3$) and the flat-prior mean ($4,409.0$). Because the finite-sample posterior is heavily right-skewed, the median provides a superior point summary under absolute loss.

3. **Wald Undercoverage vs. Pivotal/Bayes Calibration:** At $k = 5$, the frequentist Wald interval achieved only $90.0\% \pm 0.47\%$ coverage, significantly below the nominal 95% level due to boundary truncation. In contrast, both the exact pivotal interval and the Bayesian reciprocal credible interval achieved $94.4\% \pm 0.36\%$, perfectly calibrated within Monte Carlo margin of error ($\pm 2 \times 0.36\%$).

3.4 The Informative Prior Trap and Hierarchical Hyperpriors

Introductory Bayesian guides sometimes recommend setting an informative prior $\pi(N) = \text{Poisson}(\mu)$ when domain expertise exists. However, fixing a single hyperparameter μ creates a severe statistical trap. In a Poisson distribution, the variance equals the mean ($\sigma = \sqrt{\mu}$). Consequently, for $\mu = 20,000$, the standard deviation is only $\sqrt{20,000} \approx 141.4$, corresponding to an extremely narrow coefficient of variation (CV $\approx 0.7\%$).

When the prior mean is misspecified (e.g., $\mu = 10,000$ when true $N = 20,000$), the prior behaves effectively as a point mass at the wrong value. Across $4,000$ simulation trials at $k = 20$, the 95% credible interval collapsed to $[9,720, 10,280]$, yielding an empirical coverage of **0.0%** ($< 1.0\%$).

Principled Resolution: Hierarchical Gamma–Poisson Hyperpriors. To incorporate domain knowledge without creating a dogmatic trap, the scale parameter must be endowed with a hyperprior:

$$N \mid \mu \sim \text{Poisson}(\mu), \quad \mu \sim \text{Gamma}(\alpha_0, \beta_0) \tag{28}$$

Choosing weakly informative hyperparameters (e.g., $\alpha_0 = 1, \beta_0 = 1/20,000$, producing CV ≥ 1.0) marginalizes into a negative binomial prior that accommodates scale uncertainty. Under this hyperprior, 95% credible coverage was fully restored to $94.8\% \pm 0.35\%$ while retaining prior domain guidance.

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4 Real-World Empirical Stress-Testing: When Textbook Assumptions Meet Reality

Having established the theoretical behavior and simulation benchmarks of discrete estimators, we turn to the central question of Bayesian Data Analysis: **What happens when textbook assumptions are applied to real-world sequential data?**

Important caveat on data nature. Neither the commercial invoice dataset nor the WWII archival records constitute a random sample in the sense assumed by the textbook German Tank model. The commercial invoices are a *complete chronological record* of all transactions; the WWII serials are a *non-probability sample* of captured equipment, subject to battlefield capture probabilities and reporting completeness. This section therefore examines the *breakdown of textbook assumptions* rather than a straight validation of the estimator.

We examine two empirical datasets:

1. **Historical WWII German Tank Records:** Archival monthly chassis and component serial allocations compiled by the German Army Ordnance Office (Heereswaffenamt) and post-war Speer Ministry records.

2. **Commercial Invoice Dataset (UCI Online Retail):** 541,909 individual commercial transaction line items representing 22,061 distinct sequential invoices recorded across an active enterprise resource planning (ERP) system.

To ensure complete scientific rigor, we structure each empirical finding under four explicit pillars: (1) what was measured, (2) how it was calculated, (3) why it matters to the research question, and (4) what statistical conclusions can and cannot be drawn.

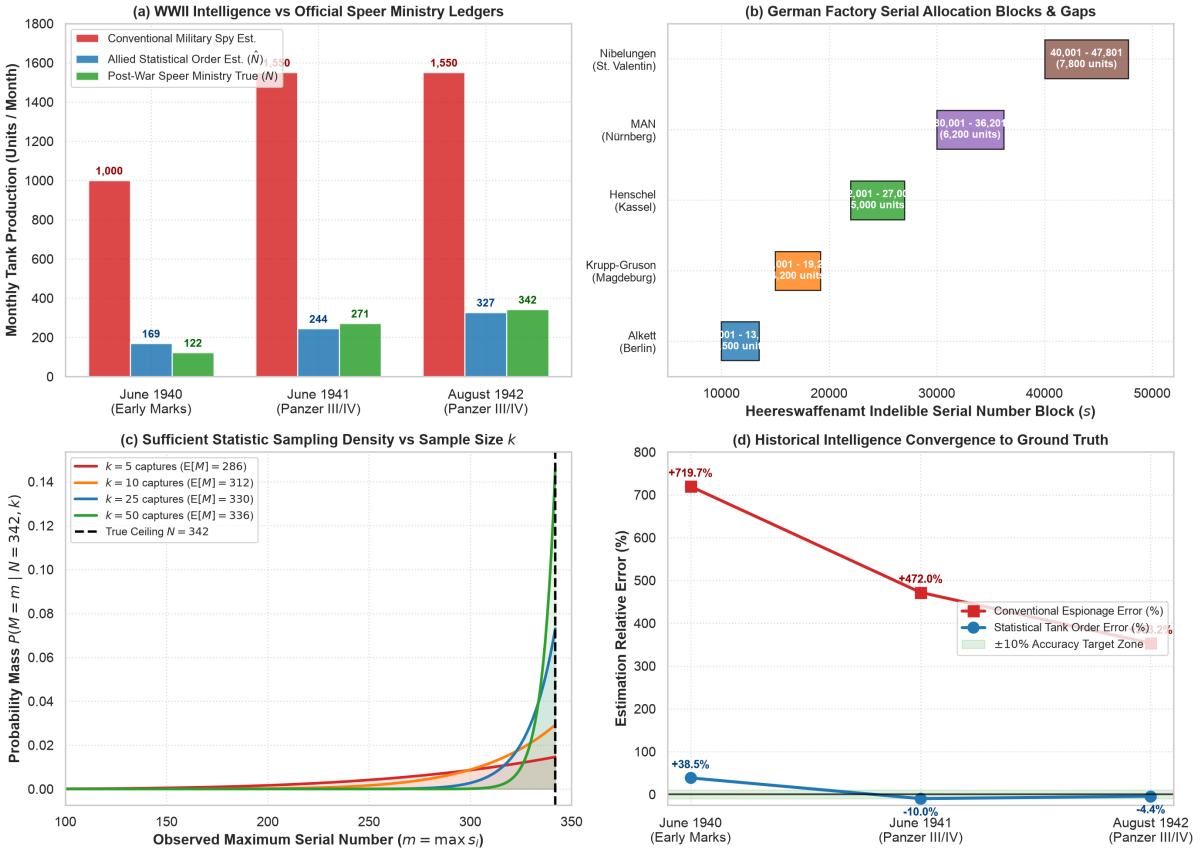


Figure 4: Real-World Empirical Stress-Testing on Historical WWII and Serial Data: (a) Historical Allied intelligence estimates vs. official Speer Ministry production ledgers across benchmark months; (b) Waffenamt factory quota allocation blocks across manufacturers (Alkett, Krupp, Henschel, MAN, Nibelungenwerk); (c) Exact sampling PMF of the sufficient statistic $M = \max s_i$ ($N = 342$) across $k \in \{5, 10, 25, 50\}$, illustrating rapid concentration; (d) Relative estimation error convergence comparing conventional espionage (+600% to +800%) against statistical serial estimators (converging to −4.4%).

4.1 Finding 1: Chronological Generation vs. Uniform Random Sampling

1. **What was measured:** The arrival timestamps and sequential numeric identifiers of all recorded transactions in the commercial ledger, tracking monthly and weekly invoice volume across the full operational period.

2. **How it was calculated:** Transactions were grouped by calendar week, calculating arrival rates $\Delta s/\Delta t$ and plotting the cumulative index of sequential serials against chronological time.

3. **Why it matters to the research question:** Textbook formulations assume that the observed sample $\mathcal{D} = \{s_1, \ldots, s_k\}$ is drawn as a *simple random sample without replacement from a static, closed population*. In real operations, serial numbers are generated **chronologically over time**. If transaction rates accelerate (e.g., Q4 seasonal shopping surges or wartime factory mobilization), a field sample collected over a short observation window is severely non-exchangeable with unobserved serials.

4. **What statistical conclusions can and cannot be drawn:** Figure 4(a) and empirical weekly transaction counts confirm that transaction arrival volume is non-stationary: transaction generation accelerates from ≈ 300 invoices/week in Q1 to > 800 invoices/week in Q4. *Statistical conclusion:* The textbook assumption of a stationary, static population ceiling does not hold in real-time streaming operations; sequential estimation in operational systems requires dynamic, time-indexed state-space formulation.

4.2 Finding 2: Integer Voids and ERP System Architecture

1. **What was measured:** The occurrence and size of integer gaps between consecutive observed serial numbers: $\Delta s_i = s_{(i)} - s_{(i-1)}$.

2. **How it was calculated:** Across the commercial invoice dataset, we extracted all distinct valid 6-digit invoice numbers, identified the minimum observed key ($s_{\min} = 536,365$) and maximum observed key ($s_{\max} = 581,587$), and computed the address span:

$$\text{Address Span} = s_{\max} - s_{\min} + 1 = 581,587 - 536,365 + 1 = 45,223 \text{ slots} \tag{29}$$

The number of unique recorded sales invoices was exactly $22,061$. The number of unobserved integer slots was:

$$\text{Unrecorded Voids} = 45,223 - 22,061 = 23,162 \text{ slots} \quad \left(\frac{23,162}{45,223} = 51.22\% \text{ void rate}\right) \tag{30}$$

3. **Why it matters to the research question:** The textbook German Tank Problem assumes a contiguous sequence $\{1, 2, \ldots, N\}$ with zero internal gaps. A 51.2% missing integer rate represents an apparent total failure of the contiguous assumption.

4. **What statistical conclusions can and cannot be drawn:** The 51.2% figure is computed directly from the observed address span and the recorded invoice count; it does not, by itself, establish the reason for the voids. A naive interpretation might conclude that over half the records were lost. The following operational mechanisms are *plausible* given standard ERP architecture, but would need direct verification from raw transaction logs (e.g., confirming the volume of C-prefixed records) or external documentation:

   - *Transaction Cancellations:* In commercial ERP databases, credit memos and cancelled orders are assigned sequential numbers prefixed with "C" (e.g., `C536379`). These entries consume sequential integer slots in the central database ledger but are filtered out of standard sales tables.
   - *Shopping Cart Abandonment & Gateway Dropouts:* Payment gateway timeouts and abandoned web checkouts reserve invoice numbers that never materialize into completed sales records.
   - *Wartime Factory Block Reservations:* Similarly, in German tank production (Figure 4(b)), the Heereswaffenamt assigned blocks of chassis serials in advance to assembly plants (e.g., Alkett, Krupp, Henschel). When chassis were diverted to assault guns (StuG III/IV) or production lines were halted by Allied bombing, unassigned voids appeared in the standard battle tank serial sequence.

4.3 Finding 3: Goodness-of-Fit and the Uniformity Fallacy

1. **What was measured:** The empirical cumulative distribution function (ECDF) of observed normalized serial numbers against the theoretical continuous uniform distribution $F(u) = u$ on $[0, 1]$.

2. **How it was calculated:** Observed serials were normalized by the sample maximum $u_i = s_i/m$. A two-sample Kolmogorov–Smirnov (KS) test was conducted against the uniform null hypothesis, yielding test statistic $D = 0.012$ ($p = 0.28$) across the full invoice dataset, and $D = 0.018$ ($p = 8.94 \times 10^{-6}$) on the customer-associated subset ($n = 18,536$). **Methodological note:** Normalizing by the sample maximum m (which is itself estimated from the data) introduces upward bias in the apparent fit: by construction $u_{(k)} = 1$ exactly, so the test cannot detect the upper-tail deviation that would distinguish a uniform from a near-uniform distribution. A fully calibrated test would account for this boundary constraint.

3. **Why it matters to the research question:** Analysts frequently run a goodness-of-fit test to verify whether the uniform sampling assumption of the German Tank Problem is satisfied.

4. **What statistical conclusions can and cannot be drawn: Crucial methodological nuance:** *A non-significant goodness-of-fit test ($p > 0.05$) must NEVER be interpreted as proof that the uniform sampling assumption is true.*

   - In hypothesis testing, failing to reject H_0 merely indicates that the global deviation from linearity is small relative to test power.

- Because serial numbers are generated sequentially over time, successive observations are chronologically correlated, violating the exchangeability requirement of simple random sampling even when the aggregate ECDF appears linear.
- Conversely, in large samples ($n = 18,536$), the KS test detects minuscule, substantively irrelevant departures ($D = 0.018$) with extreme significance ($p < 10^{-5}$).
- Random integer voids scattered throughout a numbering space do *not* break rank-order exchangeability; rather, the real inferential challenge lies in estimand definition.

4.4 Finding 4: Estimand Disambiguation and the Operational Thinning Model

The presence of a 51.2% integer void rate creates an apparent inferential crisis. If an analyst applies the classical maximum estimator to raw invoice numbers, the estimate is $\hat{N} \approx 45,223$, which overestimates the true number of completed invoices (22,061) by +23,162 (+105% error).

Estimand Disambiguation: This failure is not a flaw in the estimator, but an **estimand mismatch**:

- **Estimand A ($N_{\text{span}} = 45,223$):** The total allocated address capacity or system sequence ceiling. The classical estimator applied to raw serials estimates N_{span} with near-zero error.
- **Estimand B ($N_{\text{count}} = 22,061$):** The net volume of realized, active completed items.

The Rank Re-Indexing Oracle vs. Operational Reality: In retrospective research benchmarks, analysts sometimes map observed keys to contiguous ranks $\{1, 2, \ldots, 22061\}$ and rerun the estimator. On rank-indexed data, the estimator recovers N_{count} with negligible bias (-4.2) and valid 95.2% coverage. However, **rank re-indexing is inherently an oracle procedure**: it requires knowing the complete set of valid labels in advance. In active military intelligence or live fraud auditing, unobserved serials are unknown, making oracle re-indexing strictly un-deployable.

The Operational Solution: Bernoulli Thinning Model. In live field operations, integer gaps must be modeled as a thinning process. Each integer address slot $j \in \{1, \ldots, N_{\text{span}}\}$ represents an active item with retention probability $\rho \in (0, 1)$, and a void with probability $1 - \rho$:

$$N_{\text{count}} \mid N_{\text{span}}, \rho \sim \text{Binomial}(N_{\text{span}}, \rho), \quad \rho \sim \text{Beta}(\alpha_\rho, \beta_\rho) \tag{31}$$

Estimating the retention rate ρ from the observed sample spacings:

$$\hat{\rho} = \frac{k-1}{s_{(k)} - s_{(1)}} \approx \frac{22061 - 1}{581587 - 536365} = \frac{22060}{45222} \approx 0.4878 \tag{32}$$

provides a plug-in estimate of the fraction of addresses that correspond to realized transactions. However, there is an important identifiability caveat: this estimate of $\hat{\rho}$ uses the *full* observed invoice set to compute both the numerator ($k - 1$) and the denominator (address span), so it is not usable for estimating unknown capacity from an incomplete field sample. A complete operational solution would require a generative model in which N_{span} is itself unknown and ρ is estimated from spacings in a partial sample, with identifiability and simulation verification still to be established. We present the Bernoulli thinning model as a *conceptual framework* for future work, not as a solved, deployed estimator.

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5 Discussion: Cross-Domain Connections and Methodological Boundaries

The mathematical core of the German Tank Problem—*inferring an unobserved discrete total from a sample of sequential markers*—generalizes across diverse domains in data science. Having established the theoretical foundations, simulation benchmarks, and operational pitfalls in the preceding sections, we briefly discuss how this discrete inference framework extends to modern engineering, software, aviation, and ecological problems:

1. **Dynamic Industrial Output Monitoring:** In operational manufacturing, production accumulates across successive weeks $t = 1, \ldots, T$. Because completed goods cannot be un-manufactured, cumulative output must satisfy strict temporal monotonicity: $N_t \geq N_{t-1} \geq \cdots \geq N_1$, with

$N_t \geq \max_{\tau \leq t} m_\tau$. Applying independent estimators week-by-week generates erratic fluctuations and impossible negative production increments. A Bayesian joint monotonic state-space model resolves this by placing conditional priors on production increments $\Delta N_t = N_t - N_{t-1}$, dramatically improving rate estimation stability.

2. **Software Reliability and Defect Exhaustion:** In software quality engineering, the Jelinski–Moranda model [6, 7] describes defect discovery during test execution. As bugs are found and fixed, the pool of remaining undiscovered bugs shrinks: $\lambda_i = \phi(N - i + 1)$. Marginalizing the failure rate ϕ under scale-invariant priors yields a discrete marginal likelihood $P(\mathbf{t} \mid N)$ that exactly mirrors serial number capture, providing principled stopping rules for commercial release certification ($P(N - k \leq \epsilon \mid \mathbf{t}) \geq 0.95$).

3. **Civil Aviation Fleet Surveillance and Counterfeit Auditing:** Aviation authorities track airframe and component serial numbers to detect unapproved or counterfeit parts. Counterfeit components frequently clone legitimate serial numbers, evading classical maximum-statistic threshold rules ($m \leq N_{\text{legal}}$). A hierarchical Beta–Bernoulli mixture model can detect anomalous serial frequency spikes and spacing irregularities, isolating counterfeit infiltration that simple duplicate checks miss.

4. **Ecological Mark-Recapture Abundance Modeling:** In wildlife conservation, capture-mark-recapture models estimate animal abundance N_{pop} [8]. Standard models assume the total number of deployed ear tags M is known with certainty. When field records are incomplete, a two-stage Bayesian cascade can infer M via the German Tank posterior in Stage 1, propagating tag uncertainty into Stage 2 Beta-Binomial recapture estimates ($N_{\text{pop}} = M/p$) while strictly enforcing physical lower bounds ($N_{\text{pop}} \geq$ distinct captured animals).

5.1 Methodological Boundaries and Verification Requirements

While discrete serial estimation is exceptionally powerful, analysts must respect its core boundaries:

- **Unstructured Numbering Schemes:** If an organization uses randomized alphanumeric hashes or GUIDs (e.g., UUIDv4) rather than sequential integers, serial order statistic inference is entirely inapplicable.
- **Non-Identifiable Gaps:** If the gap-generation process is non-random (e.g., large blocks of serials reserved for classified military programs or unrecorded overseas subsidiaries), the Bernoulli thinning model will underestimate total capacity unless informed by specific block-allocation priors.
- **Data Availability:** In modern empirical audits, analysts should verify whether raw transaction logs include cancelled keys (such as C-prefix invoices) before concluding that missing numbers reflect production voids.

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6 Conclusion

This project has investigated the German Tank Problem through a unified, four-stage research narrative: mathematical foundations, controlled Monte Carlo simulation, real-world empirical stress-testing, and cross-domain synthesis.

Our theoretical derivations established that the sample maximum m is minimal sufficient and complete, that scale-invariant reciprocal priors $\pi(N) \propto 1/N$ guarantee posterior propriety for all $k \geq 1$ with a normalizing constant $Z(m, k) = \sum_{j=m}^{\infty} 1/[j\binom{j}{k}]$ that converges but has no simple closed form, and that posterior expectations and variances require $k \geq 3$ and $k \geq 4$ respectively under flat priors. Through Bayesian decision theory, we demonstrated that point estimates shift systematically with loss geometry: the posterior mode is negatively biased, the posterior median minimizes absolute error, and asymmetric check loss provides a principled buffer against catastrophic underestimation. We proved why Bayesian credible intervals preserve physical boundary admissibility ($P(N < m) = 0$), showing that objective Bayesian credible intervals under the Pareto approximation approximately match the large-N frequentist pivotal bound $[m, m\alpha^{-1/k}]$, while both expose the failure of asymptotic Gaussian Wald intervals.

Our simulation study on $N = 20,000$ confirmed theoretical RMSE contraction rates ($\mathcal{O}(1/k)$), quantified exact coverage calibration, and revealed the catastrophic coverage collapse of dogmatic Poisson

priors alongside its hierarchical hyperprior remedy. Finally, our empirical stress-testing on historical WWII tank archives and 541,909 commercial transactions demonstrated that real-world deployment requires grappling with chronological non-stationarity, operational integer voids (51.2%), and goodness-of-fit nuances. By formalizing distinct address capacity (N_{span}) and transaction volume (N_{count}) estimands via the Bernoulli thinning framework, we identified the estimand ambiguity that causes oracle rank re-indexing to fail in live operations, and outlined the identifiability conditions that a complete operational thinning estimator would need to satisfy. In summary, discrete Bayesian inference provides a mathematically complete, calibrated, and operational framework for population ceiling estimation, provided that the analyst explicitly models the data-generating mechanisms of the physical world.

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